SISS: Smart Inventory Sorting System

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Abstract

This paper presents a two-part approach. Part 1 explores the application of Object-Oriented Programming (OOP) using the Qt C++ QML Framework to address inefficiencies in logistics and supply chain management. We leverage core OOP concepts to provide a solution through the Qt framework. Part 2 validates this approach by incorporating existing statistical data that demonstrates the prevalent issues in the logistics and supply chain industry, such as labor costs and a problem/opportunity table. The Sustainable Development Goal (SDG) 9, Industry, Innovation, and Infrastructure, is highly relevant to our project as it emphasizes the need for technological advancements to drive sustainable growth in various sectors, including logistics and supply chain management. Our innovative solution integrates a digital app with sensor devices to automate processes, optimize inventory management, and improve decision-making. By democratizing access to advanced tools and fostering inclusivity, our solution aligns with the principles of SDG 9. The paper concludes by demonstrating how OOP with Qt can create a simple Minimum Viable Product (MVP) that could potentially solve the identified problems, generate new revenue streams, and serve as a foundation for future technological solutions.

Part 1

Software Overview

As the user launches the software, they are prompted to input their username and password. They enter their credentials and proceed. Now, they face a decision point: to either cancel or log in. If they choose to cancel, they are asked for confirmation - "Are you sure you want to cancel?" If they decline, they are redirected back to the login screen. If they confirm the cancellation, they are prompted with another query: "Do you want to close the app?" If they answer "no," they are redirected back to the login screen. However, if they choose to exit, the application closes. Additionally, users can sign up by entering their name, email, and password.

On the other hand, if the user opts to log in, the software verifies their credentials. If successful, the home screen will open. However, if the login fails, they receive an error message prompting them to enter valid credentials. Once logged in, the user is presented with various options on the home screen: "Activity," "Settings," and "Warehouse."

Selecting "Activity" leads the user to a section where they can manage their products. Here, they can create, edit, update and delete product details such as name, timestamp, size, section, quantity and category.

Choosing "Settings" allows the user to access their profile. Within the profile section, they can make adjustments as necessary.In the settings, they can log out when needed. If they end up choosing "Warehouse" they can view storage information here. The storage area provides insights into the storage usage within the application.

Software Flowchart

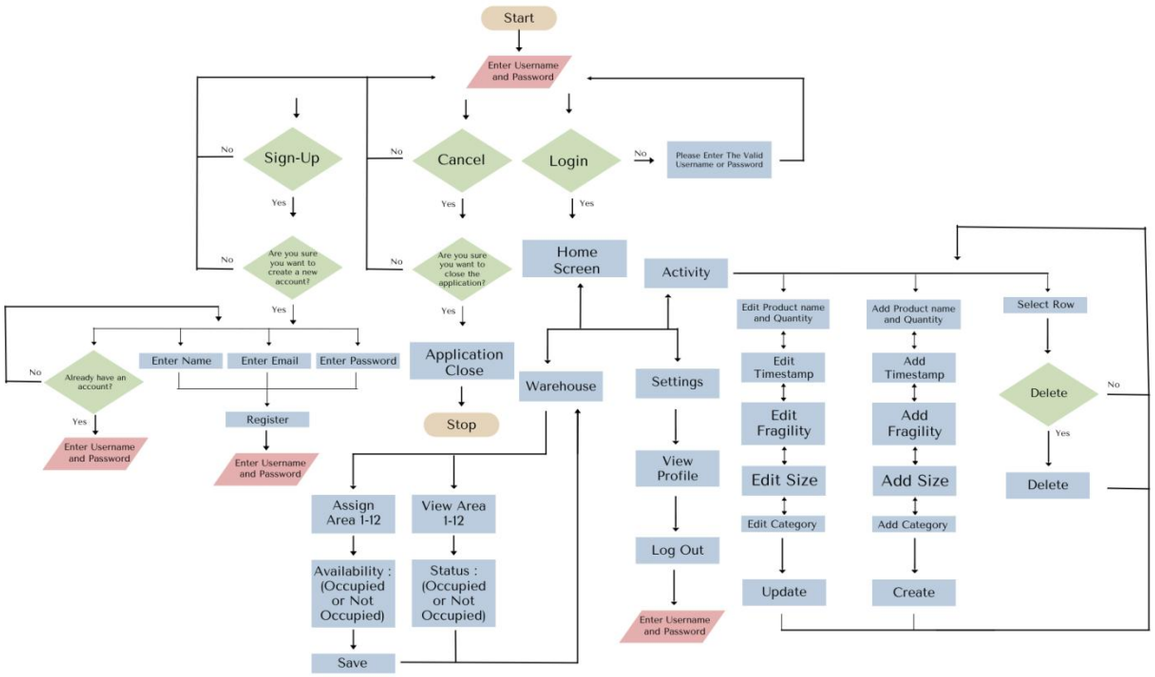


Figure 1. Software Flowchart

Software Sequential Screenshots

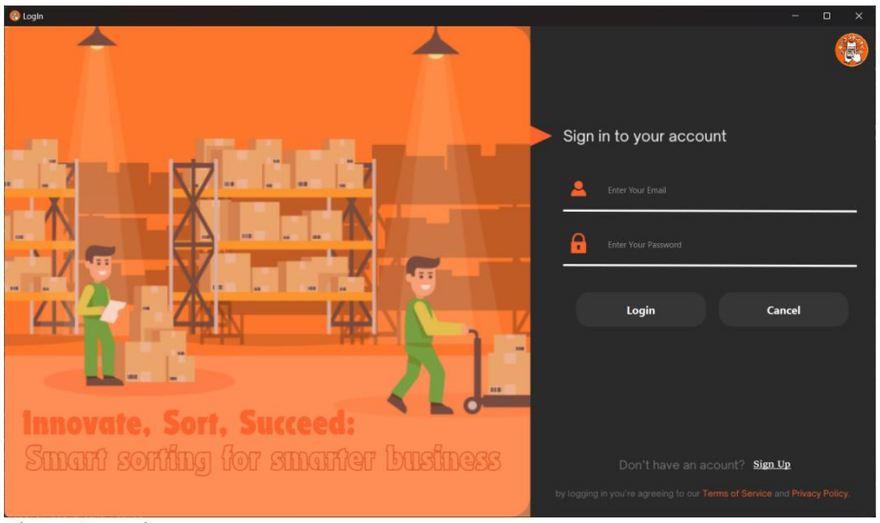


Figure 2. Login Screen

If you already have an account, you can input your email and password here. If you are a new user, you can press "Sign Up" to create an account.

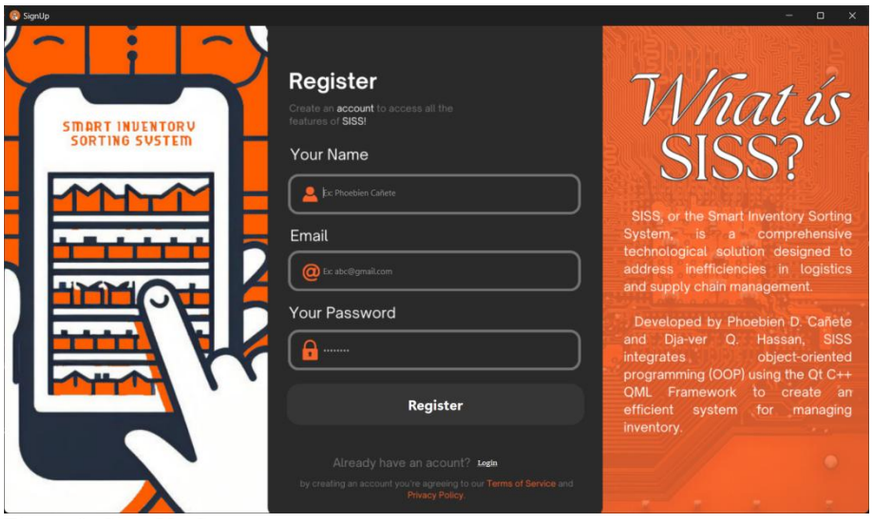


Figure 3. Sign-Up Screen

Assuming you are a new user, register here by entering your name, email, and desired password, then click "Register" to create your account. It will then automatically return to the login screen, where you can log in with your newly created account.

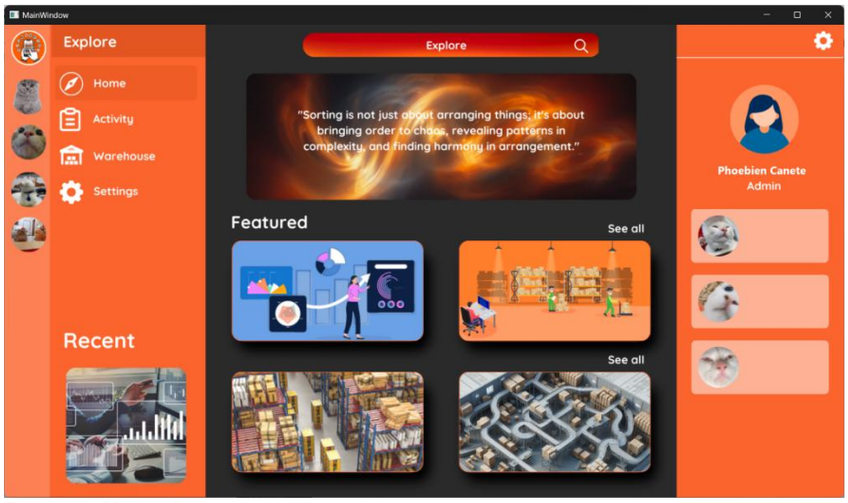


Figure 4. Home Screen

Here you can view recent data and preview your warehouse. You can also see your profile here. You can choose where to go by pressing the activity button, warehouse button, or settings button.

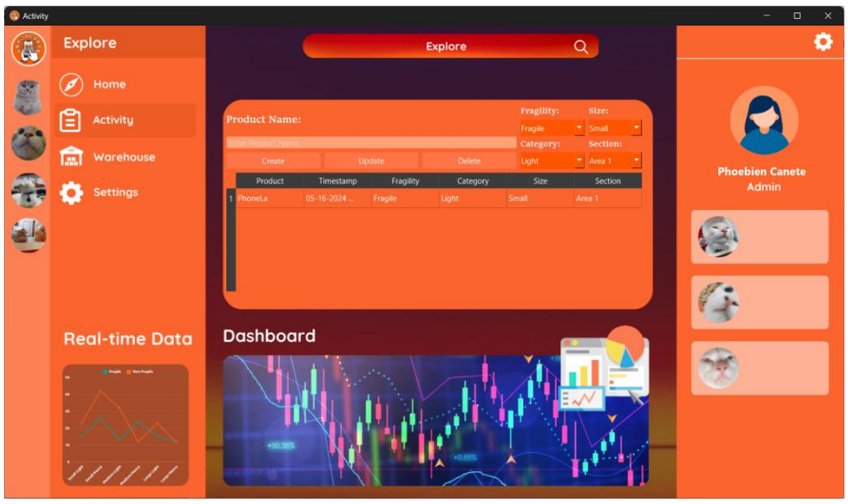


Figure 5. Activity Screen

In your activity screen, you can create, update, and delete a product you have chosen. You can also view real-time data from the warehouse. From here, you can go to the home screen, warehouse screen, and settings screen.

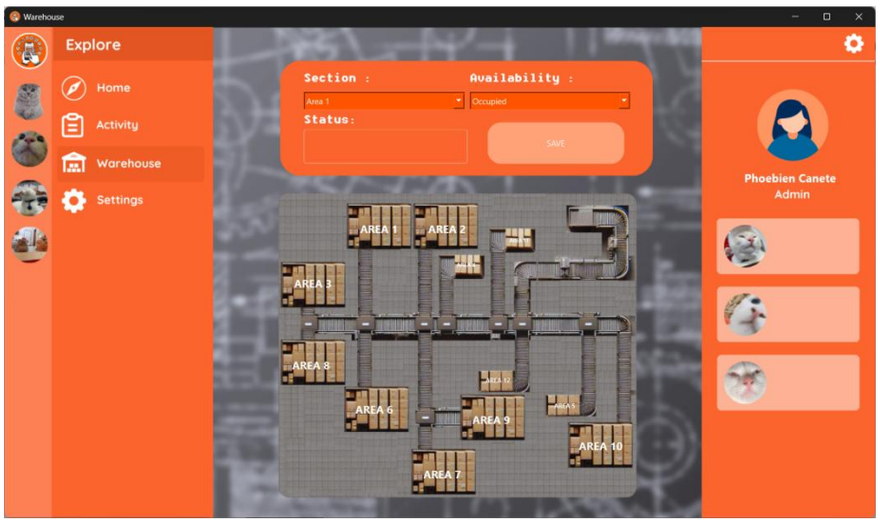


Figure 6. Warehouse Screen

In the warehouse screen, you can update the availability of the areas in the warehouse. You can also view their status by clicking on the area you want to view. From here, you can go to the home screen, activity screen, and settings screen.

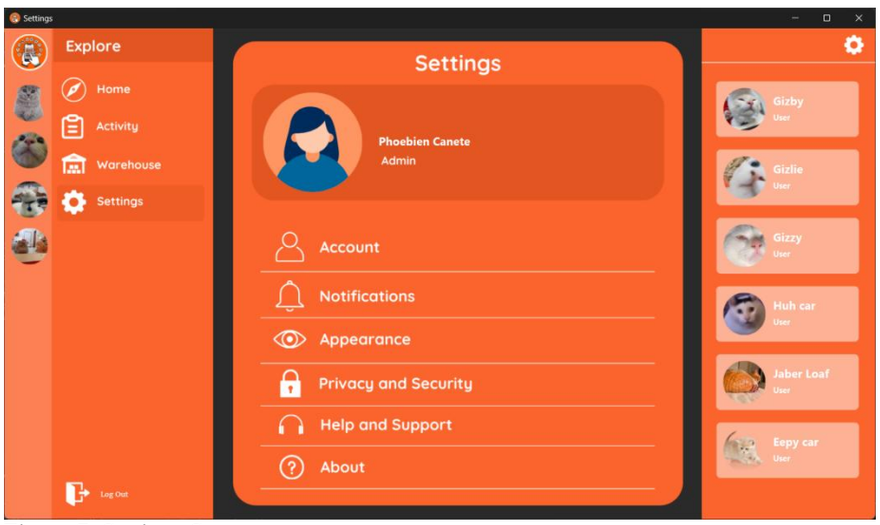


Figure 7. Settings Screen

Here, users are supposed to be able to manage their account details, such as name, email, and profile picture, ensuring their personal information is up to date. Users can customize their experience with options like notification preferences (email or in-app), appearance settings for theme and font size, and privacy/security controls, along with essential app details in the About section. If you press "Log Out," it will log out of the account and automatically return to the LogIn Screen.

Screenshot of Project Folder Library

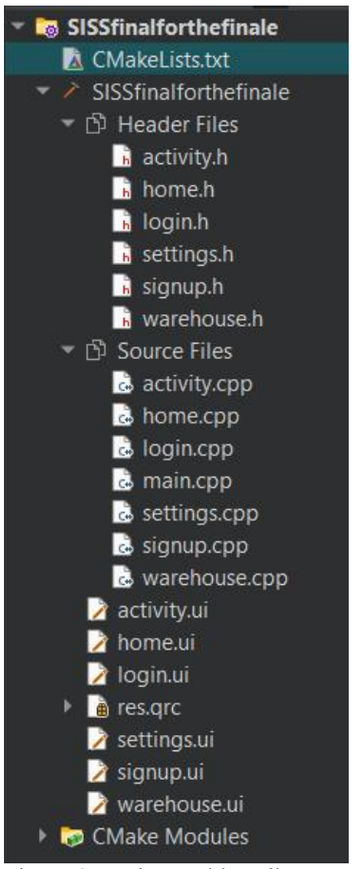


Figure 8. Project Folder Library

Screenshot of Database

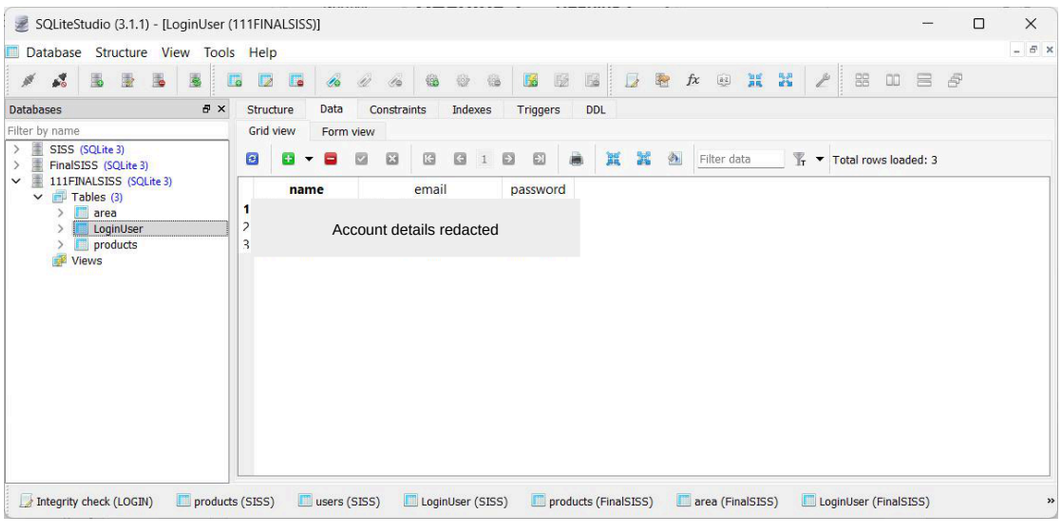


Figure 9. Database for the Sign-Up and Login with name, email and password as attributes

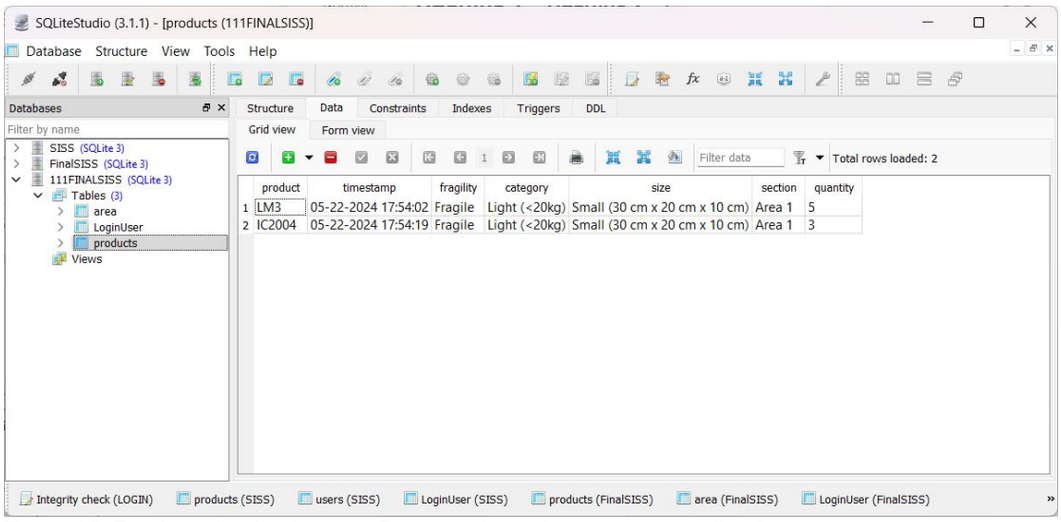


Figure 10. Database for the Activity Screen with product, timestamp, fragility, category, size, quantity and section as attributes

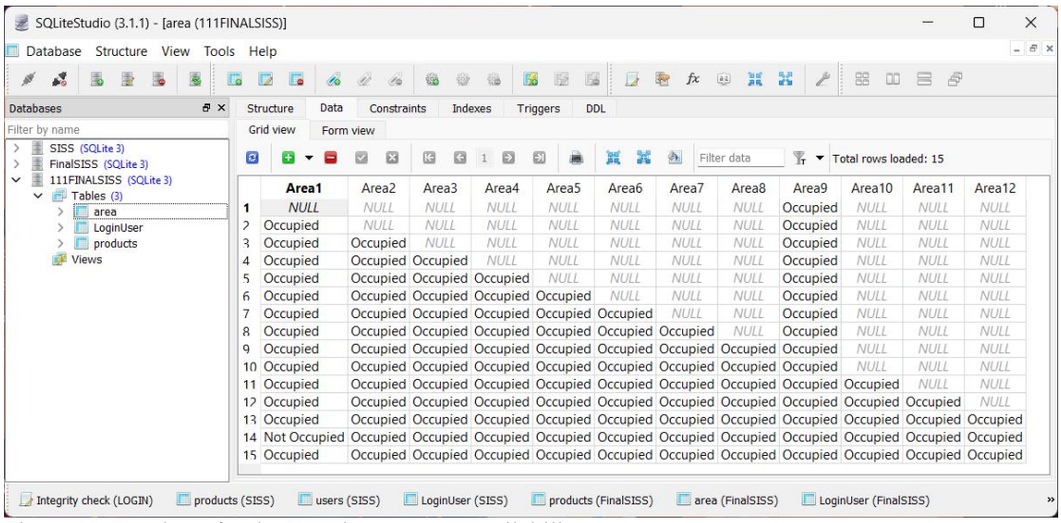


Figure 11. Database for the Warehouse Area Availability

Software Logo

Our logo is a visual representation of our System Application, known as the Smart Inventory Sorting System, signifies our dedication to the Sustainable Development Goal (SDG) theme of Industry, Innovation, and Infrastructure. At its core, the logo features a smartphone, symbolizing the modern, digital nature of our system, with a finger tapping the smartphone to signify user interaction, highlighting the system's user-friendly accessibility. Surrounding the smartphone are bubbles, each representing a unique feature or function of our comprehensive system, all accessible through the app, illustrating its wide-ranging capabilities. The smartphone symbolizes not just design but our system's purpose—to provide efficient and organized inventory management at users' fingertips.

Developer Recommendations and Conclusion

As developers, we envision several key recommendations to enhance our current system. Firstly, we propose integrating augmented reality (AR) technology into the app to provide users with immersive experiences for warehouse navigation and task management. Secondly, we want to use machine learning algorithms to enhance predictive analytics capabilities would enable more accurate forecasting of inventory trends and demand fluctuations. Additionally, expanding the use of Internet of Things (IoT) devices beyond sensors to include smart tags and RFID technology can enhance asset tracking and real-time monitoring. Moreover, developing advanced data visualization tools within the app and implementing personalized customer engagement strategies would empower users with deeper insights and foster long-term engagement. Finally, establishing a dedicated team for continuous performance optimization ensures that the system remains responsive to evolving user needs and technological advancements, solidifying its position as a leading solution for automated warehouse and inventory management.

Part 2

Overview *(Technological Solution)*

Our innovative solution addresses inefficiencies in logistics and supply chain management by integrating a digital app with sensor devices. By automating processes and optimizing inventory management, we aim to revolutionize industry practices and drive sustainable growth in line with SDG 9. Our system utilizes sensor devices to assess goods' weight, size, and condition in real-time, benefiting stakeholders such as retail stores and factories. The app, complementing sensor integration, enables seamless identification, sorting, and management of merchandise, empowering informed decision-making through data visualization features. Introducing a membership model ensures sustainable revenue, while efficiency gains lead to reduced operational costs and increased profitability. Moreover, our solution democratizes access to advanced tools, fostering inclusivity, and promoting environmental responsibility within the industry.

Keywords: Efficiency, Automation, Cost Reduction, Accessibility

Introduction

We want to change the system in the logistics and supply chain industry. To also reduce human error and increase productivity. The current situation of manual labor of managing this industry is inefficient and the lack of experience of employees further slow things down and make more mistakes ending up for more costs. We think that it is an innovative solution that combines software, technology and logistics to meet the demands of the modern market.

Problem Definition

In the logistics and supply chain industry, problems like human error, labor cost, inefficiencies, lack of experience of employees, and lower productivity are common. They slow things down, cause mistakes, and drive-up costs. The usual fixes aren't cutting it, so it's time for new, innovative solutions.

Our approach emphasizes the pivotal role of technology in addressing these challenges. Implementing innovative strategies such as automation, data analytics, and potentially artificial intelligence presents a significant opportunity for the industry to enhance operational efficiency, minimize errors, and elevate productivity levels. This fusion of technological solutions with logistical processes marks a transformative phase in how goods are transported, monitored, and managed. By embracing these advancements, businesses can not only adapt to the rapid pace of today's market but also maintain a competitive edge in the industry landscape.

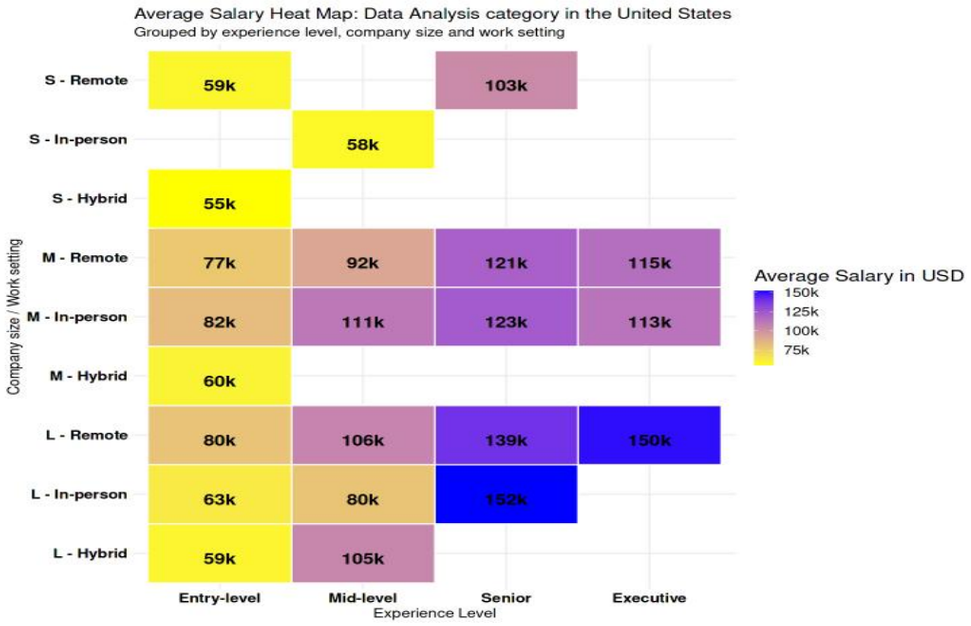


Figure 12. Recent data that shows employment costs and salaries

Problem/Opportunity

Context	Problem	Alternatives
It occurs when there is human error. When the workers take unnecessary breaks. If the available space is not correctly optimized. Labor shortages that lead to inefficiencies.	The root cause of the problem may be the significant amount of money wasted due to inconvenience, as well as the extensive labor required to organize and store various items in a limited-space warehouse or storage facility.	Stakeholders could invest in the training and development of their workers.
Stakeholders	Emotional Impact	Alternative Shortcomings
People who manage businesses, such as retail stores and malls, and possibly even small businesses who need to organize their storage efficiently. Factories that produce products and materials may also face these problems.	Stakeholders would feel frustrated by the inefficiencies and errors that disrupt operations, concerned about the financial impact of wasted resources and potential business loss, and overwhelmed by the complexity of managing large inventories and the need for constant optimization.	Training programs can be costly and time-consuming, potentially leading to temporary productivity dips; they may also cause stress for employees, be complicated to implement effectively, and risk losing newly trained employees to other opportunities.
Quantifiable Impact		
Can get inventory accuracy by percentage and reduction in labor costs measured in currency.		

Table 1. Problem/Opportunity Overview and Alternatives.

Value Proposition



Figure 13. Value Proposition

Technological Solution Overview

Our technological solution simplifies logistics and supply chain management by combining sensor devices and a digital app. The sensors assess goods in real-time, providing crucial data on weight, size, and condition. Meanwhile, the app automates processes like identification, relocating and sorting of the goods, and offers data visualization for informed decision-making. This integration enhances efficiency, accuracy, and cost-effectiveness throughout the whole process.

Technological Solution Flowchart

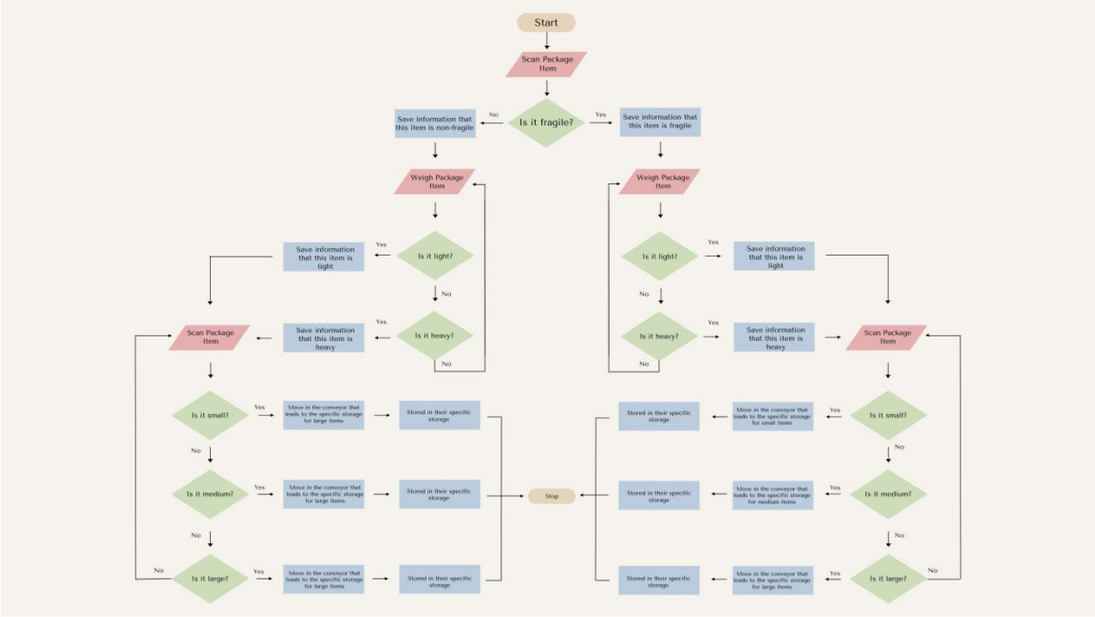


Figure 14. Technological Solution Flowchart

Technological Solution Prototype

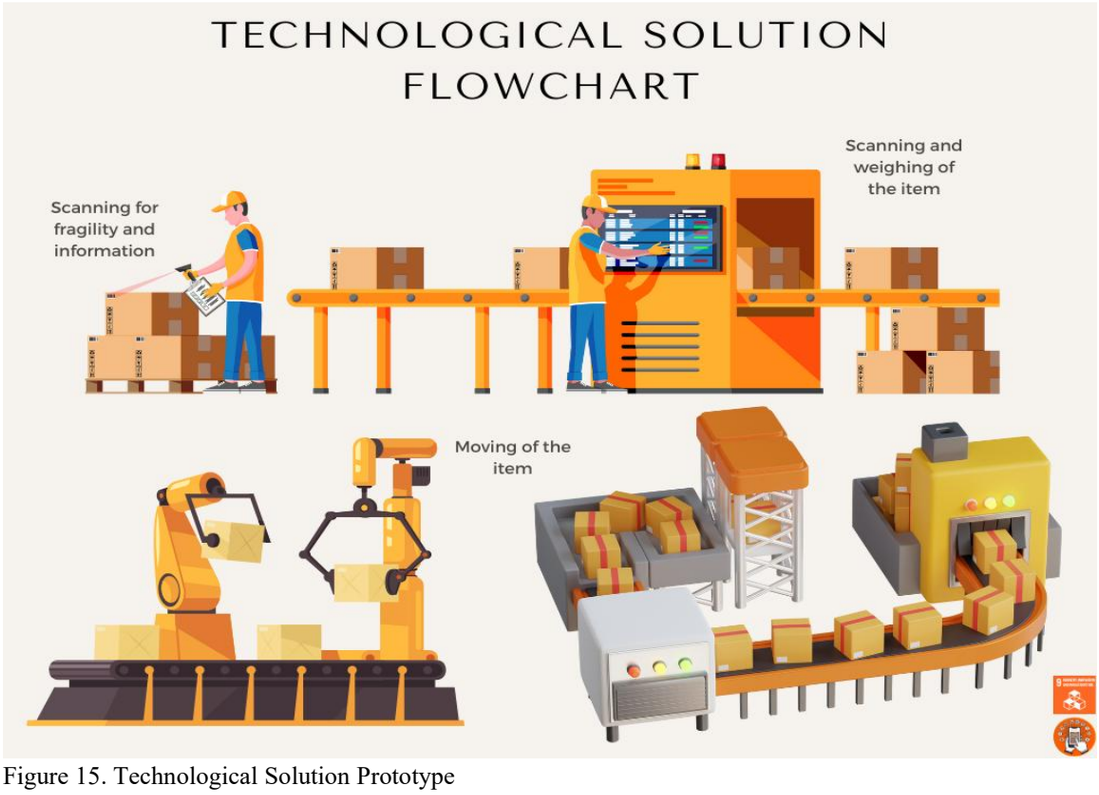


Figure 15. Technological Solution Prototype

The package will be scanned to determine if it is fragile or not. Then, the system will save the information about the package. Next, it will be weighed, and you can choose between light or heavy. The system will save that information and scan the package again to determine if it is small, medium, or large. After that, they will be stored in their specific storage.

Technological Solution Components

The Ultrasonic Sensor measures sizes, sorting items into small, medium, or large categories. The HX711 amplifier and load cells weigh goods, marking them light or heavy for efficient tracking. Meanwhile, the OV7670 Camera Module checks for fragility, ensuring delicate items are handled with care. Together, these tools will work to keep the warehouse running smoothly, ensuring every item finds its place with precision and care.

Hardware Components	Function	Corresponding Condition
Ultrasonic Sensor	This can accurately measure distances between objects, enabling the system to measure the sizes of the goods and merchandises	If item is small, medium or large, then move in the conveyor that leads to the specific storage for small, medium or the large items.
HX711 amplifier and load cells	The HX711 amplifier and load cells allows the system to accurately measure the weight of goods or merchandise. This enables precise inventory tracking, ensuring that stock levels are accurately recorded and managed.	If item is light or heavy, then save the information whether its light or heavy and proceed with the conveyor to the next process.
OV7670 Camera Module	This can enable object recognition functionality within the system. This allows for automatic categorization and sorting of inventory items based on their visual characteristics.	If item is fragile or not fragile, then save the information whether its fragile or not and proceed with the conveyor to the next process.

Table 2. Technological Solution Components

Team Composition



Phoebien D. Cañete

A senior high school graduate, she studied in the STEM strand. She's taken on various leadership roles, like being the SPG Vice-President and Page Editor for the school publication. She enjoys diverse activities such as boxing, swimming, baking, and business, showing her balanced personality. Her role as a layout artist for the school publication indicates her creativity and attention to detail.



Dja-ver Q. Hassan

He completed Senior High School, majoring in Graphic Design under Information and Communications Technology (ICT). He's into graphic design and coding, showing his dedication to his field and his eagerness to learn new things. Dja-ver stands out for his quick learning skills, letting him easily adjust to new tech and situations.

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